Linear Programming (LP)

LP P

LP in standard form: $\min c^T x$ s.t. $Ax = b, x \geq 0$

where $X \in R^q$ decision variables

and $c \in R^q$ costs vector

and $A, p \times q$ matrix of constraints

and $b \in R^p$

$I = (X, c)$ where X set of feasible solutions and c is the cost function

$X = \{x \in R^q : Ax = b, x \geq 0\}$

$c : X \to R, c(x) = c^T x$

EX

$\min x_1 + x_2$

s.t. $x_1 + 2x_2 = 3$

$x_1, x_2 \geq 0$

$$x = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$$

$$c = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$A = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$$

$$b = \begin{pmatrix} 3 \end{pmatrix}$$

$x^* = (0, 3/2)$ where this represents the optimal solution

without loss of generality (wlog) $rank(A) = p$ (full row rank which implies $q \geq p$) this is because if the rank is not p, then the last row (or few rows) are linear combinations of previous rows. This means that the last constraint (or last few constraints) are useless because it can be represented by the previous rows and no longer represents anything new. If (by error) the b does not line up for the last row(s) then the constraints make the solution impossible.

> Since $rank(A) = p, \exists p$ column vectors of A that are the basis of R^P
>
> This means that $A = (BL)$ where B is the basis vectors and L is leftover of A.

Max instead of Min (and other non standard forms)

1. $\max = -\min$
2. $x_1 \leq 0$ because $-x_1 = y_1, y_1 \geq 0$
3. Suppose $x_1 \geq$ or $< 0 \rightarrow x_1 = y_1 - z_1, y_1, z_1 \geq 0$
4. $\sum_j a_{ij}x_j - s_i = b_i$ (for inequality $\sum \geq b_i$) » $s_i \geq 0$ is the slack variable
5. $\sum_j a_{ij}x_j + s_i = b_i$ (for inequality $\sum \leq b_i$)

EX

$\max x_1 + x_2$

s.t. $x_1 + 2x_2 \geq 3$

$2x_1 + 3x_2 \leq 4$

$x_1 \leq 0$

Working Transformations:

$y_1 = -x_1 \geq 0$

$x_2 = y_2 - z_2 (y_2, z_2 \geq 0)$ (since x_2 is unbound)

For the max, $\min y_1 + z_2 - y_2$

Thus we have (rn)

$\min y_1 - y_2 + z_2$

s.t. $-y_1 + 2y_2 -z_2 -s_1 = $

$-2y_1 + 3y_2 - 3z_2 + s_2 = 4$

$y_1, y_2, z_2, s_1, s_2 \geq 0$

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$\max x_1 + x_2$

s.t. $2x_1 + 3x_2 \leq 12$

$0 \leq x_1 \leq 4$

$0 \leq x_2 \leq 3$

Plot here :)

DEF Let $x, y \in R^q (x \neq y)$

Strict convex combinations of x, y

$z = \lambda x + (1\lambda)y, \lambda \in (1, 0)$

THM If $x, y \in X$, then any strict convex combination is feasible (note this means we can get a z between x and y)

Proof: remember that $\min c^T x$, s.t. $Ax = b, x \geq 0$

$x, y \in X\{Ax = Ay = b, x, y \geq 0, Az = A(\lambda x + (1 - \lambda)y)$

$Az = \lambda Ax + (1 - \lambda)Ay$

$Az = \lambda b + (1 - \lambda)b = b$

and we know $z \geq 0$ bc its all positive components.

Thus z can be added because it has no contradictions

THM If x^*, y^* is optimal, then any strict convex combination is optimal

Proof: $x^*, y^* \in X \to z^*$ feasible

$v^* = c^T x^* = c^T y^*$

$c^T z^* = c^T($ im lazy proof is plug and simplify $) = v^*$

Def $x \in X$ is an extreme point iff it is not strict convex combinaton of $y, z \in X$ (note $\in X$ denotes feasible and this is fancy to say x is a corner because it is not between two points)

THM If LP has a finite optimal solution then $\exists$ opitmal solution extreme point(s).

This means that extreme points are candidates for optimality

Suppose we have an optimal solution x^*, the non-zero components of x^* relate to A such that those components of A are a set of lin indep vectors. In fact, since any extreme point can become the optimal solution from different cost functions, all edge points $\bar{x}$ can create a set of lin indep vectors from A via the same method above.

THM Let $\bar{x}$ be a feasible solution

$A = [a_1, a_2, ..., a_q]$

Define $\mathcal{A}_c = \{a_j : \bar{x}_j > 0\}$ (note this is a caligraphic A)

Extreme point iff $\mathcal{A}_c$ is a set of lin indep vectors

Proof 1 See examples

Proof 2 See correctness of simplex algorithm (later)

Since $rank(A) = p, \exists$ Basis of R^p among $a_1, ..., a_q$

Def Basis of A

$\bar{x}$ feasible $\mathcal{A}_c$

If $\mathcal{A}_c$ is a basis of A then $\bar{x}$ extreme pt (because if $\mathcal{A}_c$ is basis then its columns are lin indep)

If $\bar{x}$ is extreme pt, then $\mathcal{A}_c$ basis? No because we don't know if $\mathcal{A}_c$ contains enough vectors to create a basis of R^p. However we can complete $\mathcal{A}_c$ to be a basis B of A

Do all basis of A correspond to an extreme point? Suppose we have a basis B and leftover L from A. x_B basic decision vectors and x_L non-basic decision vectors.

$Ax = b \to Bx_B + Lx_L = b$

$\to x_B B^{-1} L x_L = B^{-1} b$ (we know B^{-1} exists because B is a basis)

If we want an extreme point then $\bar{x}_B = B^{-1} b$

Def A basic solution corresponding to a basis B is $\bar{x} =$ [B^{-1}b, 0]$

$B \to \bar{x} \to \mathcal{A}_c \subset B \to A$ lin indep $\bar{x}$ is extreme pt

However is $\bar{x}$ feasible? NO bc we cannot guarentee $\bar{x} \geq 0$

Def Basic Feasible Solution (bfs) is a basic soolution and feasible so $x_B \geq 0$

We find that if our basis results in a feasible solution then $\bar{x}$ is an extreme point

LP in Canonical Form

$\min(c_L^\pi)^T x_L (+\pi^T b)$ [note the constant term can be ommitted]

s.t. $x_b + \bar{A}x_L = \bar{b}$

$x_b, x_l \geq 0$

$\bar{A} = (\bar{a}_{i,j})\bar{b} = (\bar{b}_i)$

EX

$\min 3x_3 - x_4$

s.t. $x_1 - 3x_3 + 3x_4 = 6$

$x_2 - 8x_3 + 4x_4 = 4$

$x_1, x_2, x_3, x_4 \geq 0$

BFS is $\bar{x} = (6, 4, 0, 0)$

NOT OPTIMAL bc Cost has a minus

THM Let (1) be LP in canonical form and $\bar{x}$ corresponding BFS

$\bar{x}$ is opt iff $c^\pi \geq 0$

Remark: optimality condition

Proof Non-degenerate case ($\delta > 0$)

$\leftarrow$ If $c^\pi \geq 0$, then $z = (c_L^\pi)^T x_L \geq 0$ (by constraints)

Meanwhile, bfs $\bar{x}, z = 0 \times \bar{x}_B + (c_L^\pi)^T \times 0 = 0 \rightarrow \bar{x}$ optimal

$\rightarrow$ Assume $c_s^\pi < 0$. Let

$\delta = \bar{a}_{i,s} \forall i \rightarrow \infty$ otherwise $\min_{h:\bar{a}(hs)>0} \frac{\bar{b}_h}{\bar{a}_{hs}}$

From $\bar{x}$ create x' under the following conditions:

$x'_j = 0$ if $j \in L - \{s\}$, δ if $j = s$, $\bar{x}_j - a_{js}\delta$ if $j \in B$ [note s is the variable we are trying to increase]

1) x' feasible

$x_i + \bar{a}_{is}x_s + \sum \bar{a}_{ij}x_j = \bar{b}_i$

For $x'_i : \bar{b}_i - \bar{a}_{is}\delta + a_{is}\delta + 0 = \bar{b}_{ij}$ and we know $\delta > 0$ because assuming non degenerate case

Show $x'_j \geq 0$ demonstrated fairly straightforward because $\delta > 0$ except the case $j \in B$

$x'_j = \bar{b}_j - \bar{a}_{js}\delta$, if $\bar{a}_{js} \leq 0$ then clearly positive (double negative), if $\bar{a}_{js} > 0, \bar{b}_j \bar{a}_{js}\delta = \bar{b}_j - \bar{a}_{js} \min_{h:\bar{a}(hs)>0} \frac{\bar{b}_h}{\bar{a}_{hs}}$ we can sub $\frac{\bar{b}_j}{\bar{a}_{js}}$ in because it is greater than or equal to the min which results in the equation being 0 Thus clearly $\bar{x}'_j \geq 0$

2) x' has lower cost:

$z' = (c^\pi_L)^T x_L = c^\pi_s \times \delta < 0$ because assuming $c^\pi_s < 0$

Proof Degenerate case ($\delta = 0$) ends up revisiting the same spot over and over while changing Basis which map to the same extreme point

Lexigraphical order of B to prevent going back to same basis